

Rescaling used by AUTO code for extensible strip simulations

BASILE AUDOLY

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Parameters from real experiment:

- L strip length
- a width
- t thickness
- ρg force per unit length
- ω_i strain

Define an *arbitrary* reference length $\mathcal{L}$

Parameters passed to AUTO code:

- $\bar{L} = \text{parL:par\#5} = L / \mathcal{L}$
- $\bar{\rho g} = \text{rhoG:par\#4} = \text{parRhoGOverEIBar} = \frac{\rho g}{\frac{Y a t^3}{12 \mathcal{L}^3}}$
- $\text{ASq:par\#2} = \text{parASqrOverT} = \frac{a^2}{\mathcal{L} t}$

Internally, AUTO uses

- $\bar{S} \in (0, 1)$
- $\bar{\omega}_i = L \omega_i$ (strain)
- $\text{aSqrOverT} = \frac{a^2}{\mathcal{L} t} / \bar{L} = \frac{a^2}{L t}$
- $\text{lambda} = \text{aSqrOverT} \sqrt{12(1 - \nu^2)} = \frac{a^2}{L t} \sqrt{12(1 - \nu^2)} = \frac{1}{\omega^* L}$
- $\bar{m}_i = \frac{m_i}{\frac{Y a t^3}{12 L}}$ (internal moment)
- a scaled energy (see mathematica file used for automatic code generation)

$$\bar{W} = \frac{W}{\frac{Y a t^3}{12 L^2}} = \frac{1}{2} \bar{\omega}_2^2 + \frac{1}{2} \frac{2}{1 + \nu} \bar{\omega}_3^2 + 3 \text{aSqrOverT} (\nu \bar{\omega}_2^2 + \bar{\omega}_3^2)^2 \varphi(\text{lambda} \bar{\omega}_2)$$

- $\bar{r} = \frac{r}{L}$ implying $\frac{r}{\mathcal{L}} = \bar{L} \bar{r}$ (position)
- $\bar{n} = \frac{n}{\frac{Y a t^3}{12 L^2}}$ (internal force)

1. Touch-down length

We measure the touch-down length $L = L_{\text{td}}$ from experiments, and set $\mathcal{L} = L_{\text{td}}$, i.e., $\bar{L} = 1$. We run auto with the default value $\bar{L} = 1$ and with $\text{parASqrOverT} = \frac{a^2}{\mathcal{L} t} = \frac{a^2}{L_{\text{td}} t}$. In AUTO, we do a continuation over $\bar{\rho g}$ to determine the special value $\bar{\rho g}_{\text{td}}$ corresponding to touch-down (see mathematica notebook).

2. Torsional buckling instability

We change the definition of the reference length, $\mathcal{L} = 270$ mm. Therefore,

$$\bar{\rho g} = \frac{\rho g}{\frac{Y a t^3}{12 \mathcal{L}^3}} = \frac{\rho g}{\frac{Y a t^3}{12 L_{\text{td}}^3}} \left(\frac{\mathcal{L}}{L_{\text{td}}} \right)^3 = \bar{\rho g}_{\text{td}} \left(\frac{270 \text{ mm}}{L_{\text{td}}} \right)^3.$$